

New York University Tandon School of Engineering
Computer Science and Engineering
CS-GY 9223I: Midterm Practice.

Concepts to Know

Random variables and concentration.

- Linearity of expectation and variance.
- Indicator random variables and how to use them.
- Markov's inequality, Chebyshev's inequality (should know from memory).
- Union bound (should know from memory).
- Chernoff and Bernstein bounds (don't need to memorize the exact bounds, but can apply if given).
- General idea of law of large numbers and central limit theorem.
- The probability that a normal random variables $\mathcal{N}(0, \sigma^2)$ falls further than $k\sigma$ away from its expectation is $\leq O(e^{-k^2/2})$.

Hashing, Dimensionality Reduction, High Dimensional Vectors

- Random hash functions.
- Random hashing for load balancing.
- Random hashing for distinct elements estimation.
- What's a streaming algorithm? What's a sketching algorithm?
- Locality sensitive hash functions.
- MinHash and SimHash for Jaccard Similarity and Cosine Similarity.
- S-curve tuning.
- Statement of Johnson-Lindenstrauss lemma (know from memory).
- Statement of *distributional* JL lemma and how it can be used to prove JL.

Convex optimization

- Gradient descent, stochastic gradient descent, and online gradient descent algorithms.
- Definition(s) of convex function.
- Why is gradient descent a "steepest decent" method?
- When can stochastic gradient descent be faster than gradient descent?
- Definitions of G -Lipschitz, β -smooth, α -strongly convex.
- How do these properties effect the convergence of gradient descent?
- Be able to compute gradients of basic functions from $\mathbb{R}^d \rightarrow \mathbb{R}$.
- How much time does it take to multiply an $n \times d$ matrix by a $d \times m$ matrix?

Practice Problems

Random variables and concentration.

From *Foundations of Data Science* book (<https://www.cs.cornell.edu/jeh/book.pdf>):

- **Exercises:** 2.1, 2.5, 2.9, 2.28, 2.29, 2.28, 2.36, 2.41, 6.6, 6.10.
1. Show that for any random variable X , $\mathbb{E}[X^2] \geq \mathbb{E}[X]^2$.
 2. Show that for independent X and Y with $\mathbb{E}[X] = \mathbb{E}[Y] = 0$, $\text{var}[X \cdot Y] = \text{var}[X] \cdot \text{var}[Y]$.
 3. Indicate whether each of the following statements is **always** true, **sometimes** true, or **never** true. Provide a short justification for your choice.
 - (a) $\Pr[X = s \text{ and } Y = t] > \Pr[X = s]$. ALWAYS SOMETIMES NEVER
 - (b) $\Pr[X = s \text{ or } Y = t] \leq \Pr[X = s] + \Pr[Y = t]$. ALWAYS SOMETIMES NEVER
 - (c) $\Pr[X = s \text{ and } Y = t] = \Pr[X = s] \cdot \Pr[Y = t]$. ALWAYS SOMETIMES NEVER

Hashing, Dimensionality Reduction, High Dimensional Vectors

1. SimHash can be applied to binary vectors $y, q \in \{0, 1\}^d$. How does doing so compare to MinHash? How is the Jaccard similarity $J(y, q)$ related to the cosine similarity $\cos(\theta(y, q))$?
2. Suppose there is some unknown vector μ . We receive noise perturbed random samples of the form $\mathbf{Y}_1 = \mu + \mathbf{X}_1, \dots, \mathbf{Y}_k = \mu + \mathbf{X}_k$ where each $\mathbf{X}_i$ is a random vector with each of its entries distributed as an independent random normal $\mathcal{N}(0, 1)$. From our samples $\mathbf{Y}_1, \dots, \mathbf{Y}_k$ we hope to estimate μ by $\tilde{\mu} = \frac{1}{k} \sum_{i=1}^k \mathbf{Y}_i$.
 - (a) How many samples k do we require so that $\max |\mu - \tilde{\mu}| \leq \epsilon$ with probability 9/10?
 - (b) How many samples k do we require so that $\|\mu - \tilde{\mu}\|_2 \leq \epsilon$ with probability 9/10?
3. Let Π be a random Johnson-Lindenstrauss matrix (e.g. scaled random Gaussians) with $O(\log(1/\delta)/\epsilon^2)$ rows. Prove that with probability $(1 - \delta)$,

$$\min_{\mathbf{x}} \|\Pi \mathbf{A} \mathbf{x} - \Pi \mathbf{b}\|_2^2 \leq (1 + \epsilon) \min_{\mathbf{x}} \|\mathbf{A} \mathbf{x} - \mathbf{b}\|_2^2$$

Is the following also true with high probability?

$$(1 - \epsilon) \min_{\mathbf{x}} \|\mathbf{A} \mathbf{x} - \mathbf{b}\|_2^2 \leq \min_{\mathbf{x}} \|\Pi \mathbf{A} \mathbf{x} - \Pi \mathbf{b}\|_2^2$$

Convex optimization

From *Convex Optimization* book (https://web.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf):

- **Exercises:** 3.7, 3.10, 3.11 (first part), 3.21 (lots of other problems if you want more practice, but many are on the harder side)
1. Let $f_1(x), \dots, f_n(x)$ be β -smooth convex functions and let $g(x) = \frac{1}{n} \sum_{i=1}^n f_i(x)$ be their average. Show that g is β -smooth.
 2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a β -smooth, α -strongly convex function. Let $g(x) = f(c \cdot x)$ for some constant $0 < c < 1$. How does g 's smoothness and strong convexity compare to that of f ? How about g 's condition number?
 3. Let $f(x) = x^4$. Is f G -Lipschitz for finite G ? Is f β -smooth for finite B ?